

RESEARCH PROFILE

I am a mathematical physicist working at the interface of quantum field theory and string theory with several branches of modern mathematics, including algebraic geometry, homological algebra, number theory, and topology. With V. Fantini, I am developing the framework of modular resurgence, a research program connecting resurgence, arithmetic, and quantum modularity, with novel implications for non-perturbative quantum physics and quantum topology.

EMPLOYMENT

2024 – present Postdoctoral Researcher in Mathematical Physics
Institut des Hautes Études Scientifiques (IHES), France
 Mentored by **Prof. Maxim Kontsevich** and funded by the **Huawei Young Talents Program**

EDUCATION

2020 – 2024 PhD in Mathematical Physics — *Très Bien (highest honours)*
 Department of Theoretical Physics, **Université de Genève**, Switzerland
 Thesis: *On the arithmetic of resurgent topological strings*
 Supervised by **Prof. Marcos Mariño** and funded by the **ERC SyG ReNewQuantum**

2018 – 2019 MSc in Mathematical and Theoretical Physics — *Distinction (highest honours)*
 Mathematical Institute and Department of Physics, **University of Oxford**, UK
 Thesis: *An introduction to motivic amplitudes*
 Supervised by **Prof. Francis Brown** and funded by the **Torno Subito Scholarship**

2015 – 2018 BSc in Physics — *Summa cum Laude (highest honours)*
 Department of Physics, **Sapienza Università di Roma**, Italy

PUBLICATIONS

PREPRINTS

- [2] *Modular resurgent structures for vectors*, with M. C. N. Cheng, I. Coman, and V. Fantini [[arXiv: 2608.08902](#)]
- [1] *Modular resurgent structures*, with V. Fantini, submitted [[arXiv:2404.11550](#)]

REFEREED RESEARCH ARTICLES

- [7] *Modular resurgence, q -Pochhammer symbols, and quantum operators from mirror curves*, with V. Fantini, **Lett. Math. Phys.** 116, 39 (2026), <https://doi.org/10.1007/s11005-026-02063-x> [[arXiv:2506.08265](#)]
- [6] *On the structure of wave functions in complex Chern–Simons theory*, with M. Mariño, **SIGMA** 22 (2026), 002, 45 pages, <https://doi.org/10.3842/SIGMA.2026.002> [[arXiv:2312.00624](#)]
- [5] *Strong-weak symmetry and quantum modularity of resurgent topological strings on local $\mathbb{P}^2$* , with V. Fantini, **Commun. Num. Theor. Phys.** 19 (2025), no. 1, pp. 1-73, <https://dx.doi.org/10.4310/CNTP.250215002033> [[arXiv:2404.10695](#)]
- [4] *Resurgence, Stokes constants, and arithmetic functions in topological string theory*, **Commun. Num. Theor. Phys.** 17 (2023), no. 3, pp. 709-820, <https://dx.doi.org/10.4310/CNTP.2023.v17.n3.a4> [[arXiv:2212.10606](#)]
- [3] *Searching for muonphilic dark sectors with proton beams*, with B. Döbrich and T.-T. Yu, **Phys. Rev. D** 106 (2022) 3, 035023, <https://doi.org/10.1103/PhysRevD.106.035023> [[arXiv:2205.09870](#)]
- [2] *An introduction to motivic Feynman integrals*, **SIGMA** 17 (2021), 032, 56 pages, <https://doi.org/10.3842/SIGMA.2021.032> [[arXiv:2009.00426](#)]
- [1] *Characterization and performance of PADME's Cherenkov-based small-angle calorimeter*, with A. Frankenthal et al., **Nucl. Instrum. Meth. A** 919 (2019) 89-97, <https://doi.org/10.1016/j.nima.2018.12.035> [[arXiv:1809.10840](#)]

PROCEEDINGS

Mini-Workshop: Resurgence, Difference Equations and Quantum Modularity, with M. Alim, V. Fantini, and L. Hollands, **Oberwolfach Rep.** 22 (2025), no. 4, pp. 2593-2632 (main reporter), <https://doi.org/10.4171/owr/2025/48>

DOCTORAL THESIS

On the arithmetic of resurgent topological strings, Doctoral Thesis Sc. 5852, **Université de Genève** (2024), <https://dx.doi.org/10.13097/archive-ouverte/unige:181979>

TALKS AND SEMINARS

2027, Jan	(*) Conference on Resurgence Theory in Mathematical Physics, TSIMF, Sanya, China
2026, Nov	(*) Mathematical Physics Seminar, LPTHE, France
Oct	(*) Workshop on Cosmology and High-Energy Physics, L2C, Université de Montpellier, France
Mar	Workshop on Mathematical Structures of Amplitudes, ESI, Austria
2025, Nov	Asymptotics, Geometry, and Physics Seminar, University of Southampton, UK
Oct	Mini-Workshop on Resurgence, Difference Equations, and Quantum Modularity, MFO, Germany
Aug	ReNewQuantum Closing Conference, SDU, Denmark
May	Random Matrix Theory Seminar, University of Oxford, UK
May	Mathematical Physics and Algebraic Geometry Seminar, CMSA, Harvard University, USA
Apr	Conference on Higher Structures, Moduli Spaces and Integrability, Universität Hamburg, Germany
Mar	Workshop on The Arithmetic of Calabi-Yau Manifolds, MITP, Germany
Jan	Youngsters Physics and Number Theory, MITP, Germany
2024, Nov	The Seed Seminar of Mathematics and Physics, IHP, France
Oct	Workshop on Holonomic Techniques for Feynman Integrals, MPP, Germany
Sep	GAP XIX: Moduli Spaces and Higher Structures, Sapienza Università di Roma, Italy
May	String Theory Seminar, Université de Genève, Switzerland
May	Workshop on Resurgence and Modularity in QFT and String Theory, GGI, Italy
May	Mathematics Seminar, Yale University, USA
Apr	String Theory Seminar, CERN, Switzerland
Mar	Seminar on Quantum Modularity and Resurgence, IHES, France
Feb	Workshop on Positive Geometry in Particle Physics and Cosmology, MPI MiS, Germany
2023, Dec	Geometry and Physics Seminar, University of Sheffield, UK
Nov	QM Research Seminar, Centre for Quantum Mathematics, SDU, Denmark
Jul	29 th Nordic Congress of Mathematicians with EMS, Aalborg, Denmark
Apr	Mathematics Seminar, IHES, France
Mar	Physical Mathematics Seminar, Université de Genève, Switzerland
2022, Nov	Workshop on Mathematics of Beyond All-Orders Phenomena, University of Cambridge, UK
2019, Nov	Seminar on Lie Groups and Moduli Spaces, University of Geneva, Switzerland
Aug	Conference on Representation Theory and Integrable Systems, ETH Zürich, Switzerland
2017, Dec	PADME Weekly Seminar, INFN – LNF, Italy

AWARDS AND SCHOLARSHIPS

2024 – 2027	Huawei Young Talents Program, IHES, France
2019 – 2020	Excellence Fellowship, NCCR SwissMAP, Switzerland
2019	Degree Prize for Distinction, St John's College, University of Oxford, UK
2018 – 2019	Torno Subito Scholarship, Department of Education, Research, and University, Lazio, Italy
2018	Best Student in Nuclear and Subnuclear Physics, Sapienza Università di Roma and INFN, Italy
2017	Summer Student Scholarship, INFN, Italy
2016 – 2018	Excellence Program, Department of Physics, Sapienza Università di Roma, Italy
2015 – 2018	Deserving Student Scholarship, Sapienza Università di Roma, Italy

TEACHING EXPERIENCE

2024, Jan	Supervision of exercise sessions on Feynman Integrals and Number Theory Winter School in Mathematical Physics, SwissMAP Research Station, Switzerland
2019, Oct	Lectures on Topological Surfaces Master Class in Mathematical Physics, Université de Genève, Switzerland
2018, Mar – May	Lectures on Riemannian Geometry Excellence Program in Physics, Sapienza Università di Roma, Italy

SCIENTIFIC ACTIVITIES

COMMUNITY ENGAGEMENT

2026 – present	Co-organizer of the Monthly Women Scientists Lunch at IHES, France
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PROFESSIONAL SERVICE

2026	Co-organizer of a one-day workshop at Laboratoire Mathématique Orsay, France
2026 – present	Referee for Communications in Number Theory and Physics
2025	Co-organizer of a five-day workshop at Mathematisches Forschungsinstitut Oberwolfach, Germany
2020 – 2024	Junior Member of the Scientific Council of the SwissMAP Research Station, Switzerland

RESEARCH NETWORKS

2024 – present	Associated Researcher, ERC SyG UNIVERSE+
2020 – 2024	Doctoral Student, ERC SyG ReNewQuantum
2019 – 2024	Excellence Fellow and Doctoral Student, NCCR SwissMAP

RESEARCH INTERSHIPS

2020	Research Internship in High-Energy Physics Phenomenology CERN, Switzerland
2019 – 2020	Master Class in Mathematical Physics NCCR SwissMAP and Université de Genève, Switzerland
2017	Research Internship in Experimental Particle Physics INFN – LNF, Italy

SKILLS

Languages	Italian (native), English, French
Programming Languages	C, C++, Python, R
Data Analysis	MATLAB, ROOT, GNU Plot
Symbolic Calculus Languages	Mathematica, SageMath, PARI/GP
Version Control Systems	Git
Simulation Software	Geant4